

Sample Math Document for Docling Testing

1 Basic Algebra¹

The quadratic formula is:²

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
³ (1)

2 Calculus⁴

The definition of a derivative:⁵

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
⁶ (2)

A definite integral:⁷

$$\int_0^\pi \sin(x) dx = 2$$
⁸ (3)

3 Linear Algebra⁹

A matrix and its determinant:¹⁰

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det(A) = ad - bc$$
¹¹ (4)

4 Probability¹²

The normal distribution probability density function:¹³

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$
¹⁴ (5)

5 Series¹⁵

Euler's identity and an infinite series:¹⁶

$$e^{i\pi} + 1 = 0$$
¹⁷ (6)

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$
¹⁸ (7)

6 Inline Math¹⁹

Here is some text with an inline formula, $E = mc^2$, embedded directly in a sentence. Another inline example: the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ appears frequently in nature.²⁰